

ZEPING LIU

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RESEARCH SUMMARY

Third-year Ph.D. student at UT Austin and **Amazon AI PhD Fellow** in **Geospatial AI**: foundation models and spatial representation learning across imagery, locations, vector geometries, and trajectories, and LLM agents that reason about space and self-improve from feedback. First-author work at NeurIPS, ISPRS P&RS, IJGIS, and IEEE TGRS; industry research at **Esri** and **Amazon**.

EDUCATION

The University of Texas at Austin — Ph.D. in Geography (Geospatial AI) 2024 – Present
Advisor: Gengchen Mai · GPA 4.00/4.00

Beijing Normal University — M.S. in Cartography and Geographic Information Engineering 2021 – 2024
GPA 3.80/4.00

INDUSTRY EXPERIENCE

Applied Scientist Intern, Geospatial Science Team, Amazon Jun 2026 – Sep 2026

- **LLM agent memory**: Designed Reflexion-style memory for an internal delivery-point quality-check agent: verified LLM reasoning against expert labels from Amazon's Last Mile Analytics and Quality (LMAQ) organization and automatically distilled lessons from past successes and failures into memory, with human-in-the-loop curation. Reached 93% accuracy using 27% of the manual effort (+10% over baseline); shortlisted by Amazon Last Mile for continued development in 2027.
- **Counterfactual Memory Calibration (CMC)**: Proposed an algorithm that makes LLM self-learning robust to incorrect labels. *Submitted to AMLC, Amazon's internal machine learning conference.*
- **StraTune (first author)**: Proposed adaptive revision-operator selection for self-evolving LLM skills; outperformed five baselines in most settings across four benchmarks, with learned skills transferring from smaller to stronger LLMs. *Under review.*
- *Performance evaluation: Inclined for a return offer.*

Software Development Intern, Spatial Statistics Team, Esri May 2025 – Nov 2025

- Built **Spatial Co-Scientist**, a multi-agent LLM system integrated with ArcGIS Pro that automates geospatial analysis via retrieval-augmented generation over Esri internal documents and tool calling, with guided analysis flows informed by expert interviews; designed a spatial reasoning mechanism that improved LLM accuracy by up to 10%.
- **Spatial CoT (IJGIS 2026, first author)**: Proposed a spatial-concept-guided chain-of-thought framework that improves LLM reasoning on complex geospatial question answering.

RESEARCH EXPERIENCE

Ph.D. Research, SEAI Lab, The University of Texas at Austin Aug 2024 – Present

- **PolyTopoBench (NeurIPS 2026 E&D, first author)**: Built a benchmark for generating complete vector polygons, including interior rings (holes), from remote sensing imagery: 4 tasks with ~300K polygons and 42K+ interior rings, 11 baselines under a unified runner, and ring-aware metrics that score region, boundary, and ring structure separately. Showed that direct vector decoders collapse on holed polygons and even the strongest models rarely recover full ring structure.
- **GAIR (ISPRS P&RS 2026, first author)**: Built a location-aware multimodal geospatial foundation model that aligns satellite and street-view imagery with geolocation through implicit neural representations; pre-trained on ~10M samples and achieved state-of-the-art results on geospatial benchmarks.
- **TrajNIA (first author, with the Google Mobility AI team)**: Developed a neural implicit attention model for continuous, scale-agnostic vehicle trajectory imputation; a dual-branch Taylor-expansion decoder with gated fusion reaches state-of-the-art accuracy with 60% fewer parameters than the trajectory foundation model UniTraj. *Under review.*
- **LocDiff (NeurIPS 2025, second author)** and **TorchSpatial (NeurIPS 2024 D&B)**: Co-developed a diffusion-based image geolocalization model in Hilbert space and a PyTorch location-encoding framework and benchmark.

Master's Research, Beijing Normal University 2021 – 2024

- Released **CBRA** (ESSD 2023, first author), the first multi-year (2016–2021) 2.5 m building rooftop dataset covering all of China, using a spatiotemporal super-resolution segmentation framework that extracts 2.5 m buildings from 10 m Sentinel-2 imagery.
- Developed building outline delineation models that combine CNN–RNN sequence prediction with line-segment and boundary cues (IEEE TGRS 2022, first author).

SELECTED PUBLICATIONS

Google Scholar: **519 citations** · **h-index 9** · **i10-index 9**

- [arXiv'26] **StraTune: Adaptive Selection of Revision Operators for Self-Evolving LLM Skills.**
Z. Liu, Y. Li, N. Lao, G. Wolff, G. Mai. *arXiv:2609.32886*, 2026 (under review).
- [NeurIPS'26] **PolyTopoBench: A Benchmark for Complex Vector Polygon Generation from Remote Sensing Imagery.**
Z. Liu, N. Lao, W. Sun, G. Wolff, Y. Xie, L. Zhao, J. Jiao, G. Mai. *NeurIPS 2026, Evaluations & Datasets Track*.
- [IJGIS'26] **Spatial CoT: A Spatial Concept Transformation Guided LLM Reasoning Framework for Complex Geospatial Question Answering.**
Z. Liu, H. Xu, N. Lao, L. Bennett, L. Zhao, M. Raad, H. Shao, L. Buie, M. Mueller, C.-C. Huang, G. Mai. *International Journal of Geographical Information Science*, 2026.
- [ISPRS P&RS'26] **GAIR: Location-aware Self-supervised Contrastive Pre-training with Geo-aligned Implicit Representations.**
Z. Liu, N. Lao, Z. Wang, J. Jiao, G. Mai. *ISPRS Journal of Photogrammetry and Remote Sensing*, 2026.
- [NeurIPS'25] **LocDiff: Identifying Locations on Earth by Diffusing in the Hilbert Space.**
Z. Wang, **Z. Liu**, J. Zhang, Z. Zhou, Q. Cao, N. Wu, L. Mu, Y. Song, Y. Xie, N. Lao, G. Mai. *NeurIPS 2025*.
- [JAG'25] **Towards the Next Generation of Geospatial Artificial Intelligence.**
G. Mai, Y. Xie, X. Jia, N. Lao, J. Rao, Q. Zhu, **Z. Liu**, Y.-Y. Chiang, J. Jiao. *International Journal of Applied Earth Observation and Geoinformation*, 2025.
- [NeurIPS'24] **TorchSpatial: A Location Encoding Framework and Benchmark for Spatial Representation Learning.**
N. Wu, Q. Cao, Z. Wang, **Z. Liu**, Y. Qi, J. Zhang, ..., N. Lao, G. Mai. *NeurIPS 2024, Datasets & Benchmarks Track*.
- [RSE'24] **Four Seasonal Composite Sentinel-2 Images for the Large-scale Estimation of the Number of Stories in Each Individual Building.**
S. Lyu, C. Ji, **Z. Liu**, H. Tang, L. Zhang, X. Yang. *Remote Sensing of Environment*, 2024.
- [ESSD'23] **China Building Rooftop Area: The First Multi-annual (2016–2021) and High-resolution (2.5 m) Building Rooftop Area Dataset in China.**
Z. Liu, H. Tang, L. Feng, S. Lyu. *Earth System Science Data*, 2023.
- [RS'23] **Learning Sparse Geometric Features for Building Segmentation from Low-Resolution Remote-Sensing Images.**
Z. Liu, H. Tang. *Remote Sensing*, 2023.
- [TGRS'22] **Building Outline Delineation from VHR Remote Sensing Images Using a Convolutional Recurrent Neural Network Embedded with Line Segment Information.**
Z. Liu, H. Tang, W. Huang. *IEEE Transactions on Geoscience and Remote Sensing*, 2022.

HONORS & AWARDS

Amazon AI PhD Fellowship (\$140,000)	2025 – 2027
AAG International Geographic Information Funds (IGIF) Scholarship (\$1,200)	2026
UT Austin Graduate Dean's Prestigious Supplement Fellowship	2025, 2026
Second Place, UT GIS Day Student Presentation	2024
Outstanding Graduate Student of Beijing; Zhou Tingru Geography Youth Award	2024
First Prize, Graduate Academic Forum, National Remote Sensing Bulletin	2023
China National Scholarship, Ministry of Education	2022, 2023

ACADEMIC SERVICE

Conference reviewer: ACM SIGSPATIAL 2025–2026, ICLR 2025, KDD 2026–2027 (AI for Science Track)

Journal reviewer: ISPRS P&RS, JAG, Transactions in GIS, Scientific Reports, GeoInformatica, Geo-spatial Information Science, Earth Science Informatics, Geocarto International, and others

Student Director, AAG Geographic Information Science & Systems (GISS) Specialty Group 2025 – 2026

TECHNICAL SKILLS

Machine learning	Python, PyTorch; remote sensing foundation-model pre-training, self-supervised learning, multimodal learning, neural representations
LLMs & agents	self-evolving LLM agents (memory, skill learning), multi-agent systems, retrieval-augmented generation (RAG), tool calling
Geospatial AI	spatial representation learning for locations and vector geometries (points, polylines, polygons); deep learning on satellite (Sentinel-2, VHR aerial) and street-view imagery; GIS software (ArcGIS Pro, QGIS)